

1.

5.2.6 (2016 SP:17)

Which one of the following will react with acidified potassium dichromate solution to give a ketone?

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 (b) $\text{CH}_3\text{CH}_2\text{CHO}$
 (c) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
 (d) $(\text{CH}_3)_3\text{COH}$

2.

5.2.8 (2016 SP:20)

Consider the following substances.

- (i) BaSO_4 (ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 (iii) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (iv) $\text{H}_2\text{NCH}_2\text{COOH}$

Which one of the following lists the substances in order of **decreasing** solubility in water?

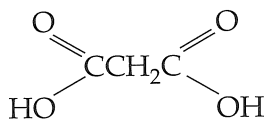
- (a) i iv ii iii
 (b) i iii ii iv
 (c) iv ii iii i
 (d) ii iv iii i

3.

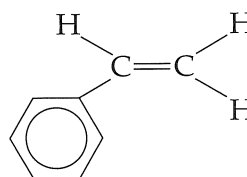
5.2.13 (2016 SP:23)

Which one of the following pairs represent monomers that could react together to form a polymer?

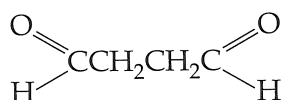
(i)



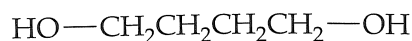
(ii)



(iii)



(iv)



- (a) i and iv
 (b) i and iii
 (c) ii and iii
 (d) iii and iv

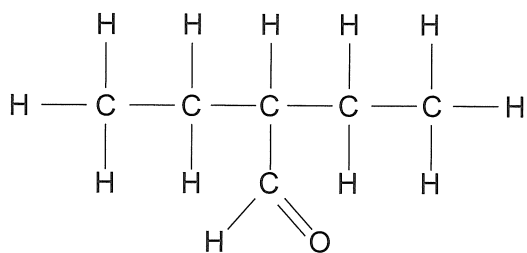
4. 5.1 (2016 SP:24)

Proteins that show a high degree of similarity in their primary structure in the Protein Data Bank are most likely to have

- (a) similar function.
- (b) identical tertiary structure.
- (c) been isolated from the same species.
- (d) the same amino acid composition.

5. 5.2.4 (2016:18)

What is the IUPAC name of the following compound?



- (a) 3-methylpentan-3-al
- (b) 2-ethylbutanal
- (c) 2,2-diethylethanal
- (d) 2-methylbutanal

6. 5.2.6 (2016:19)

Below is a table of reactions involving organic compounds.

Reaction	Product
ethene + hydrogen	1
ethanal + acidified permanganate solution	2
ethanol + acetic (ethanoic) acid with sulfuric acid	3
acetic (ethanoic) acid + sodium carbonate	4

Which row of the table below identifies a product of each reaction correctly?

	Product 1	Product 2	Product 3	Product 4
(a)	an alkane	a carboxylic acid	an aldehyde	an ester
(b)	an alkane	a carboxylic acid	an ester	carbon dioxide
(c)	an alkene	carbon dioxide	an aldehyde	a carboxylic acid
(d)	an alkane	carbon dioxide	an ester	a carboxylic acid

7.

5.2.10 (2016:20)

Which of the following compounds could be used to produce a polymer?

- (i) CH_2CHCH_3
- (ii) $\text{HOOCCH}_2\text{COOH}$
- (iii) CH_2CHOH
- (iv) HOCH_2CH_3
- (v) $\text{H}_2\text{NCH}_2\text{NH}_2$

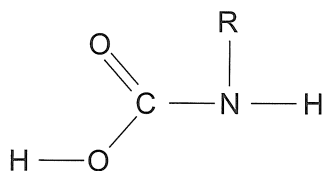
- (a) I, II, V only
- (b) I, II, IV only
- (c) I, II, III, V only
- (d) II, III, IV, V only

8.

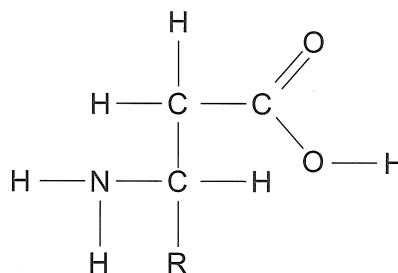
5.2.14 (2016:21)

Which of the following **best** represents the generalised structure of α -amino acids? (Note: R represents a side chain.)

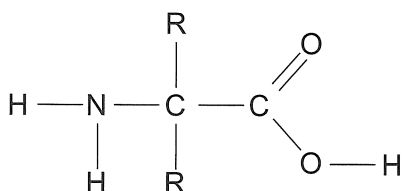
(a)



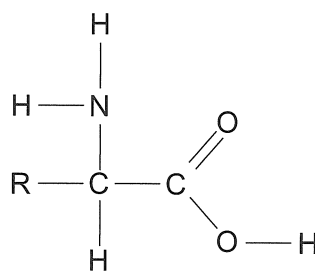
(b)



(c)



(d)



9.

5.2.5 (2017:14)

Which of the following are isomers of $\text{C}_5\text{H}_8\text{O}_2$?

- (i) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CHO}$
- (ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$
- (iii) $\text{CH}_3\text{COCH}(\text{CH}_3)\text{CHO}$
- (iv) $\text{CH}_2\text{CHCH}_2\text{CH}_2\text{COOH}$

- (a) i and ii only
- (b) i, ii and iv only
- (c) i, iii and iv only
- (d) ii, iii and iv only

10.

5.2.14 (2017:15)

Which one of the following is the dominant form of glycine in basic solution?

- (a) $\text{NH}_2\text{-CH}_2\text{-COOH}$
 (b) $\text{NH}_2\text{-CH}_2\text{-COO}^-$
 (c) $\text{NH}_3^+\text{-CH}_2\text{-COO}^-$
 (d) $\text{NH}_3^+\text{-CH}_2\text{-COOH}$

11.

5.2.3 (2017:16)

A chemist attempts to identify a pungent, colourless liquid by conducting several experiments. The results are shown in the table below:

Experiment	Observations
add acidified potassium dichromate solution	orange solution turns green
a lighted taper held above the liquid	flame and heat produced
add sodium metal	metal reacts and colourless, odourless gas evolved
add acidified, concentrated acetic (ethanoic) acid	fruity odour produced

Using this information, identify the functional group present in the liquid.

- (a) ketone
 (b) alcohol
 (c) amine
 (d) carboxylic acid

12.

5.2.17 (2017:17)

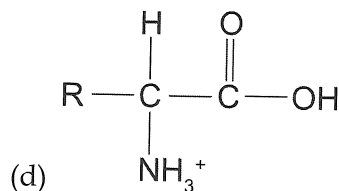
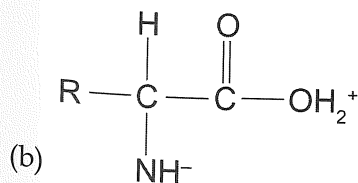
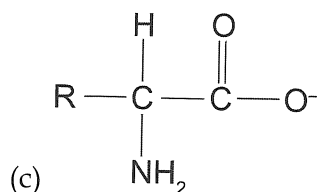
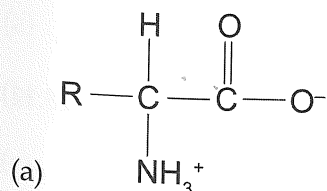
The amino acid sequence of a protein is referred to as its

- (a) primary structure.
 (b) secondary structure.
 (c) tertiary structure.
 (d) parent chain.

13.

5.2.15 (2017:19)

Which one of the following structures represents a zwitterion?



14.

5.1 (2017:20)

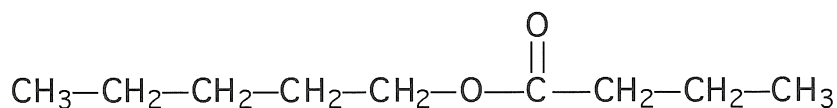
The function of a protein is linked closely to

- (a) its method of production.
- (b) the nature of its intermolecular forces.
- (c) the number of atoms bonded to it.
- (d) its structure.

15.

5.2.7 (2018:04)

The compound with the structural formula shown below smells like apricots:



Which of the following is true for this compound?

	Name of compound	Organic reactants required to synthesise this compound
(a)	pentyl butanoate	pentanol and butanoic acid
(b)	butyl pentanoate	butanol and pentanoic acid
(c)	pentyl butanoate	butanol and pentanoic acid
(d)	butyl pentanoate	pentanol and butanoic acid

16.

5.1 (2018:09)

Which of the following statements about the Protein Databank (PDB) is/are correct?

- (i) The PDB allows users to select a protein and then view its structure.
 - (ii) The PDB is updated regularly and access by scientists worldwide is free.
 - (iii) The PDB is a worldwide repository of information on all chemical substances listed in chronological order of discovery.
- (a) i only (c) i and ii only
- (b) iii only (d) ii and iii only

17.

5.2.5 (2018:11)

Which of the following molecules is capable of demonstrating *cis-trans* isomerisation?

- (a) $\text{CH}_2\text{CHCHBrCH}_3$
- (b) $\text{CH}_3\text{CHCHCH}_3$
- (c) $\text{CBr}_2\text{CHCH}_2\text{CH}_2\text{Br}$
- (d) $\text{CH}_2\text{BrCBr}_2\text{CH}_2\text{CH}_3$

18.

5.2.18 (2018:17)

Proteins can contain α -helices and/or β -pleated sheets. The intermolecular forces holding these structures in their shapes are

- (a) dispersion forces.
- (b) dipole-dipole forces.
- (c) hydrogen bonds.
- (d) ion-dipole attractions.

19.

5.2.14 (2019:09)

Which one of the following is an alpha amino acid?

(a) $\text{H}-\text{Se}-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$	(b) $\text{H}_2\text{N}-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{COOH}$
(c) $\text{H}_2\text{N}-\underset{\text{CH}-\text{COOH}}{\text{CH}_2}-\text{CH}_2-\text{COOH}$	(d) $\text{H}_2\text{N}-\underset{\text{Se}-\text{CH}_2-\text{COOH}}{\overset{\text{O}}{\text{C}}}-\text{NH}_2$

20.

5.2.6 (2019:14)

Which one of the following properties exhibited by octanol is **not** related to the dispersion forces between the molecules?

- (a) combustibility
- (b) melting point
- (c) solubility in octane
- (d) solubility in water

21.

5.2.5 (2019:15)

Which one of the following compounds will **not** exhibit *cis-trans* isomerism?

- (a) 1,2-difluoro-but-1-ene
- (b) 1,1-difluoro-but-1-ene
- (c) 1,2-difluoro-but-2-ene
- (d) 1,4-difluoro-but-2-ene

22.

5.2.6 (2019:18)

Which one of the following could **not** be a product when propan-1-ol is oxidised?

- (a) CO_2
- (b) $\text{CH}_3\text{CH}_2\text{CHO}$
- (c) $\text{CH}_3\text{CH}_2\text{COOH}$
- (d) CH_3COCH_3

23.

5.2.8 (2019:22)

Between which of the following pairs of substances can dispersion forces exist?

- (i) CH_3Cl and H_2O
- (ii) $\text{CH}_3\text{CH}_2\text{CHO}$ and HBr
- (iii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- (iv) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and NH_3

- (a) i and ii only
- (b) i, ii and iii only
- (c) iii only
- (d) i, ii, iii and iv

24.

5.2.5 (2019:23)

Which one of the following is an isomer of pentanoic acid?

- (a) $\text{CH}_3\text{CHCH}_2\text{OCH}_2\text{CHO}$
- (b) $\text{CH}_2\text{CHCH}_2\text{OCH}_2\text{CH}_2\text{OH}$
- (c) $\text{OHCCH}_2\text{CH}_2\text{CH}_2\text{CHO}$
- (d) $\text{CH}_3\text{CHCHCH}_2\text{COOH}$

25.

5.2.5 (2019:25)

How many isomers does the compound $C_2H_3Br_3$ have?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

26.

5.2.5 (2020:04)

The number of possible isomers of $C_2H_2F_2$ is

- (a) 1
- (b) 2
- (c) 3
- (d) 4

27.

5.2.1 (2020:10)

Which of these statements regarding organic molecules are correct?

- (i) Organic molecules all contain carbon.
 - (ii) Functional groups consist of groups of atoms or a particular type of bond.
 - (iii) Functional groups influence the chemical properties of organic molecules.
 - (iv) Functional groups influence the physical properties of organic molecules.
- (a) i and iii only
 - (b) ii and iv only
 - (c) i, ii and iii only
 - (d) i, ii, iii and iv

28.

5.2.15 (2020:11)

Which of the following pairs of molecules can form peptide bonds with each other?

(i)	$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{HO}-\text{C}-\text{C}-\text{C}-\text{C}-\text{OH} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$ butane-1,4-diol	$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}_2\text{N}-\text{C}-\text{C}-\text{C}-\text{C}-\text{NH}_2 \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$ butane-1,4-diamine
(ii)	$\begin{array}{c} \text{CH}_2-\text{C}_6\text{H}_4-\text{OH} \\ \\ \text{H}_2\text{N}-\text{CH}-\text{COOH} \end{array}$ tyrosine	$\begin{array}{c} \text{CH}_2-\text{C}_6\text{H}_4-\text{OH} \\ \\ \text{H}_2\text{N}-\text{CH}-\text{COOH} \end{array}$ tyrosine
(iii)	$\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_3 \\ \\ \text{H}_2\text{N}-\text{CH}-\text{COOH} \end{array}$ valine	$\begin{array}{c} \text{CH}_2-\text{C}_6\text{H}_5 \\ \\ \text{H}_2\text{N}-\text{CH}-\text{COOH} \end{array}$ phenylalanine
(iv)	$\begin{array}{cc} \text{H} & \text{H} \\ & \\ \text{H}-\text{C}-\text{C}-\text{OH} \\ & \\ \text{H} & \text{H} \end{array}$ ethanol	$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{O} \\ & & & // \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} & & & \\ & & & \backslash \\ \text{H} & \text{H} & \text{H} & \text{O}-\text{H} \end{array}$ butanoic acid

- (a) i and iv only
 (b) ii and iii only
 (c) i, ii and iii only
 (d) i, ii, iii and iv

29.

5.2.8 (2020:13)

Which of the following alcohols would you expect to have the highest boiling point?

- (a) pentan-1-ol
 (b) pentan-2-ol
 (c) pentan-3-ol
 (d) 2-methylbutan-2-ol

30.

5.1 (2020:14)

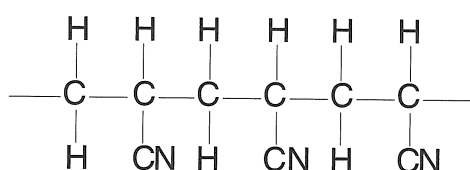
The Protein Data Bank contains information relating to the structures of proteins. The structure of a protein is important because it is related closely to its

- (a) equilibrium constant.
- (b) bonding capacity.
- (c) nutritional value.
- (d) function.

31.

5.2.11 (2020:21)

Polyacrylonitrile fibres can be used to make blankets and carpets. The structural formula of a segment of this polymer is shown below.



The structural formula of the monomer used to make polyacrylonitrile is:

(a)	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{C}-\text{H} \\ \quad \\ \text{CN} \quad \text{H} \end{array} $	(b)	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{CN} \quad \text{H} \quad \text{CN} \end{array} $
(c)	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{CN} \end{array} $	(d)	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{CN} \\ \quad \\ \text{H} \quad \text{H} \end{array} $

32.

5.22.3 (2021:02)

Chlorine gas is bubbled for several minutes through a sample of pent-1-ene. Which of the following statements identifies the type of reaction that occurs and the colour of the solution in the flask after the reaction is complete, assuming the chlorine gas is the limiting reagent?

	Type of reaction	Solution colour after complete reaction
(a)	substitution	colourless
(b)	addition	colourless
(c)	addition	green
(d)	substitution	green

33.

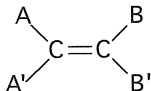
5.2.5 (2021:16)

Two isomeric forms of a saturated hydrocarbon

- (a) contain different types of atoms.
- (b) have the same structural formula.
- (c) have the same molecular formula.
- (d) react vigorously with one another.

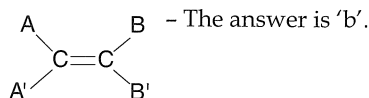
- 30.(2021:12) After consulting the Data Booklet and the Table of Reduction Potentials, the only pair with a positive E° value involves nickel and silver. The metals must be in their own solution (Ag/AgNO_3) or they will react locally. The answer is c.
- 31.(2021:13) At the anode positive charges are produced while at the cathode positive charge is destroyed. The salt bridge supplies charged ions to keep the solutions charge neutral. The answer is d.
- 32.(2021:19) In 'a' the solution would become less blue and more pale green. In 'b' if there were limited bromine it would change from orange to colourless. In 'c' the pale green chlorine would react with the colourless iodide solution and it would become brown. In situation 'd' the orange solution would remain orange. Acetic acid cannot be oxidised by acidified dichromate solution but methanoic acid can $\rightarrow \text{CO}_2$. The answer is d.
- 33.(2021:22) The syllabus requires you to know the silver example of this question. The object must be negative and reduction occurs there so it is the cathode. The anode might be pure gold to be oxidised and replenish the solution. The answer is d.

Chapter 4: Organic Chemistry

- 1.(2016 SP:17-c) Secondary alcohols can be oxidised to form ketones. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ is a secondary alcohol. The answer is 'c'.
- 2.(2016 SP:20-c) Barium sulfate is insoluble. Number (iv) can donate hydrogen bonds and accept them in two places so will be very soluble. Number (ii) can accept and donate hydrogen bonds so will be quite soluble. The answer is 'c'.
- 3.(2016 SP:23-a) To form a polymer we need a link to be formed between monomers. Substance (a) has carboxylic acid groups and could form ester linkages. So with (iv) this could form a polymer. (ii) and (iii) cannot form linkages. The answer is 'a'.
- 4.(2016 SP:24-a) The primary structure of a protein is the covalently bonded polypeptide polymer. Its 'R' groups are determined by the amino acids in the polymer. The functional groups in the 'R' groups determine the tertiary structure. If two polymers have similar functional groups they will have similar tertiary structures and hence similar shape. Function is dependent on shape. The answer is 'a'.
- The strength of the interaction between ethanol molecules and water molecules are stronger than the internal interactions of both so ethanol dissolves in water – therefore they are miscible.
- 5.(2016:18-b) The $-\text{CH}=\text{O}$ group is an aldehyde. Aldehydes always appear at the end of a chain. The base substance is therefore butanal. The answer is 'b'.
- 6.(2016:19-d) The reactions are addition to an alkene, oxidation of an aldehyde, esterification, and neutralisation acid with a carbonate. The answer is 'd'.
- 7.(2016:20-c) To make a polymer either an alkene is needed (to make an additional polymer) or the necessary groups to form a condensation polymer (polyester or polyamide). Addition is possible with I and III, while a polyester can form with II and V. The answer is 'c'.
- 8.(2016:21-d) An α -amino acid has a central α carbon a COOH , one side group (or an H) and an NH_2 . In 'a' there is no central C, 'b' has two central carbons, 'c' has two groups. The answer is 'd'.
- 9.(2017:14-c) To be isomers the molecules must have the same molecular formula. The answer is 'c'.
- 10.(2017:15-b) The question checks recognition of the fact that an amino acid contains a COOH acid group that will react with a strong base to form COO^- . The answer is 'b'.
- 11.(2017:16-b) The syllabus requires knowledge of the chemical properties of functional groups and 'these reactions can be used to identify the functional group present within the organic compound'. In turn the tests show it can be oxidised, it is flammable, it has an OH group and it is either a carboxylic acid or an alcohol. The answer is 'b'.
- 12.(2017:17-a) A question straight from the syllabus: 'the sequence of α -amino acids in a protein is called its primary structure.' The answer is 'a'.
- 13.(2017:19-a) This question is taken directly from the syllabus and identifies the acid and base properties of amino acids. The NH_2 accepts a proton and the COOH loses one. The answer is 'a'.
- 14.(2017:20-d) This question is taken directly from the syllabus. The 'lock and key' model of protein function means the shape is the essential characteristic in protein function and this is due to the structure or shape. The answer is 'd'.
- 15.(2018:04-a) When naming an ester look for the double bonded oxygen. This was part of the original carboxylic acid. The left-hand part was therefore the alcohol. So butanoic acid and pentan-1-ol were used. The substance is pentyl butanoate. The answer is 'a'.
- 16.(2018:09-c) The PDB allows users to access the structures of proteins and view them in 3D 'cartoon' mode. It is free of charge and lists proteins alphabetically. The answer is 'c'.
- 17.(2018:11-b) In an alkene cis-trans isomerism (not called geometric isomerism anymore) can occur when A and A' are different and when B and B' are different also.
- 

This only occurs in case b. The answer is 'b'.
- 18.(2018:17-c) The secondary structure is formed by hydrogen bonding between the H in a $\text{N}-\text{H}$ and an O in a $\text{C}=\text{O}$ group. Depending on the amino acid sequence, it forms either α -helices and/or β -pleated sheets. The answer is 'c'.
- 19.(2019:09-a) An alpha amino acid has an N terminus, a C terminus and one central C called the alpha carbon. There is no requirement that the correct answer be amongst those listed on the data sheet. The answer is 'a'.
- 20.(2019:14-a) Melting point and solubility are related to secondary forces. Combustion concerns breaking covalent bonds to produce activated particles. The answer is 'a'.

- 21.(2019:15-b) To exhibit cis-trans isomerism molecules must have A and A' different **and** B and B' different.



- 22.(2019:18-d) Remember that combustion is a form of oxidation and that primary alcohols can be oxidised to aldehydes and/or carboxylic acids. The answer is 'd'.
- 23.(2019:22-d) Dispersion forces are created by instantaneous dipoles caused by electrons. By definition this means all of the substances have dispersion forces. The answer is 'd'.
- 24.(2019:23-b) Pentanoic acid ($\text{C}_4\text{H}_9\text{COOH}$) has a molecular formula $\text{C}_5\text{H}_{10}\text{O}_2$. Only one distractor has H_{10} - the answer is 'b'.
- 25.(2019:25-b) $\begin{array}{c} \text{H} \quad \text{Br} \\ | \quad | \\ \text{H}-\text{C}-\text{C}-\text{Br} \\ | \quad | \\ \text{H} \quad \text{Br} \end{array}$ and $\begin{array}{c} \text{H} \quad \text{Br} \\ | \quad | \\ \text{Br}-\text{C}-\text{C}-\text{H} \\ | \quad | \\ \text{H} \quad \text{Br} \end{array}$ The answer is 'b'.
- 26.(2020:04) The molecule is an alkene so we should consider cis-trans isomers too.
1,1-difluoroethene, cis-1,2-difluoroethene, trans-1,2-difluoroethene. Answer is (c).
- 27.(2020:10) Organic molecules contain carbon. Functional groups affect the chemical and physical properties of a molecule - by e.g. alcohol demonstrating hydrogen bonds that affects boiling point and the type of alcohol determine oxidation outcomes (aldehyde, ketone). The presence of double bonds similarly changes chemical and physical properties. The answer is (d).
- 28.(2020:11) A peptide (or amide) is a condensation product of an amine and a carboxylic acid that can be used to form a polymer. The constituent molecules should have suitable functional groups either $-\text{NH}_2$ or $-\text{COOH}$ must be present. The answer is (b).
- 29.(2020:13) All of these alcohols can demonstrate hydrogen bonding with each other. Pentan-1-ol molecules can pack closest together. Therefore, they will be the hardest to separate requiring more energy that is available at a high temperature. The answer is (a).
- 30.(2020:14) The structure (shape) of a protein determines its function like a key in a lock. The answer is (d).
- 31.(2020:21) This polymer is an example of an addition polymer. The repeating unit is $\text{CH}_2=\text{CH}-\text{CN}$. The answer is (c).
- 32.(2021:02) Pent-1-ene is an alkene. They can react with halogens by addition across the double bond. If we assume the pent-1-ene is INXS then the mixture will be colourless. Dichloropentane is colourless. The answer is b.
- 33.(2021:16) Two isomers of saturated hydrocarbons could be butane and 2-methylpropane. They are both C_4H_{10} . The answer is c.

Chapter 5: Synthesis

- 1.(2016 SP:22-b) To form an ester an alcohol and a carboxylic acid are required. The ethene can form ethanol by an addition reaction in the presence of water, and the ethanol can be oxidised to form ethanoic acid. These two substances can react to form ethyl ethanoate. The answer is 'b'.
- 2.(2016:05-b) This is an equation mentioned in syllabus and should be known. NaOH produces soap while methanol produces the transesterification reaction making biodiesel. The answer is 'b'.
- 3.(2016:22-a) Hardness is a measure of calcium ion concentration. This is measured by volume of soap reacted. Other salts (chlorides etc.) will respond to silver ions or raise the boiling point but they have nothing to do with hardness. Only soap solution is indicative of hardness. The answer is 'a'.
- 4.(2016:23-b) Reaction X is a REDOX reaction with catalyst but this is not one of the answers offered. The best answer is 'b'.
- 5.(2016:24-c) This is a known reaction from the syllabus. The answer is 'c'.
- 6.(2016:25-a) This is a known reaction from the syllabus. The answer is 'a'.
- 7.(2017:11-d) There are 12 principles of green chemistry you should have read. The minimising toxic waste and by-products is essential. The answer is 'd'.
- 8.(2017:13-c) Rate determining steps have not been examined previously in the WACE. The idea is that in a sequence of reactions the overall reaction rate is dependent on the slowest step. The answer is 'c'.
- 9.(2017:21-c) The syllabus states 'the structure of the anionic detergents derived from dodecylbenzene contains a non-polar hydrocarbon chain and a sulfonate group'. You should know what a sulfonate group is. The answer is 'c'.
- 10.(2018:19-b) An enzyme lowers the activation energy so (a) and (c) are wrong. The reaction is exothermic so the answer is 'b'.
- 11.(2018:20-c) Answers (a), (b) and (d) are all **correct**. The hydration reaction does not use enzymes. The answer is 'c'.
- 12.(2018:21-b) The reverse reaction has a large E_a due to the fact the reaction is exothermic. Thus, the reverse reaction benefits more from the use of a catalyst than the forward reaction which has a relatively small E_a . The answer is 'b'.
- 13.(2018:22-d) The question asks only about yield. The reaction is exothermic thus removing heat will favour the forward reaction and the mol ratio indicates high pressure will also favour the forward reaction. The answer is 'd'.
- 14.(2018:23-a) Both reactions are exothermic and the Haber process uses a catalyst mostly made of iron while the Contact process uses vanadium pentoxide as a catalyst but remember catalysts do not alter yield - just rate. The answer is 'a'.
- 15.(2018:24-a) The long hydrocarbon chain demonstrates substantial dispersion forces with oil (another molecule with high dispersion forces). The long carbon chain has no lone pairs of electrons and so cannot engage in hydrogen bonding and since it has no ions or dipoles the answer is 'a'.
- 16.(2019:11-a) This question required detailed understanding of the two biodiesel processes and soap making along with some green chemistry. The answer is 'a'.
- 17.(2019:13-c) This question again required knowledge of the biodiesel process and required recognition of a triglyceride - (a) and (b) are products of the reaction. The answer is 'c'.